Undergraduate Physics Specialization of a 2B Language Model on a Free GPU Budget: Leak-Checked Data, Half-Strength LoRA, and Held-Out Evaluation

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***Abstract*—Small open language models are attractive for education and on-device use, yet they remain weak at undergraduate physics, and most published adaptations assume compute budgets far beyond the reach of students. We study how far a two-billion-parameter model can be specialized for physics problem solving using only free-tier NVIDIA T4 GPUs (16 GB, no bfloat16 support). We build a training set of 29,500 short, step-by-step physics solutions from public sources, removing every candidate that resembles a benchmark question and verifying each solution's final answer against its reference with the official UGPhysics grader. We fine-tune Qwen3.5-2B, a hybrid Gated DeltaNet/attention model, with LoRA in its non-thinking mode in 4.5 GPU-hours, and find that applying the learned update at half strength keeps the physics gain while reducing the loss in general mathematics. On 1,142 UGPhysics questions that were held out from every development decision, the resulting model reaches 18.9% accuracy versus 15.9% for the base model (+3.0 points, 95% interval [0.3, 5.7]; exact McNemar $p = 0.050$), its answers become 2.7$\times$ shorter, and truncations at the 4,096-token budget fall from 48.9% to 6.9%. The gain costs 9.4 points on MATH-500. We also report negative results: supervising a 1.2B always-thinking model with the same short solutions reduced its UGPhysics accuracy from 13.3% to 5.5%, and self-distillation preserved but did not improve it. Code, data, every model generation, and the model are released.**

***Index Terms*—large language models, physics reasoning, parameter-efficient fine-tuning, LoRA, benchmark contamination, low-resource training, evaluation methodology**

I. Introduction

Physics problems combine conceptual knowledge, symbolic derivation, numerical computation and unit handling, which makes them a demanding test of language-model reasoning. UGPhysics [1], a benchmark of 5,520 undergraduate problems in English across 13 subjects, showed that even 7B instruction-tuned models answer only about a quarter of the questions correctly under its official judging protocol. Models of one to two billion parameters are cheaper to run and easier to deploy for teaching, but they are weaker still, and the recipes that improve reasoning, such as distillation from large reasoning models [2] or reinforcement learning, usually require many accelerator-days.

This paper asks a practical question: *what can a single developer achieve for undergraduate physics with a 2B model and nothing but free-tier cloud GPUs?* The constraint is severe. An NVIDIA T4 has 16 GB of memory, about 65 TFLOP/s of half-precision tensor throughput, and no bfloat16 support; free sessions end after a few hours. Every design choice in this work, from the data format to the numerical precision of the released weights, follows from that budget.

We make four contributions:

- **A leak-checked, grader-verified training set** of 29,500 short physics solutions built from public data. Candidates are removed if they resemble any test question by exact text, a shared 13-word run, TF-IDF similarity, or reworded text with the same numbers, and each solution's final answer is checked against its reference with the benchmark's own grader (Sec. IV).
- **A training recipe for a hybrid-attention model on a T4**, including LoRA placement across attention, Gated DeltaNet and MLP layers, a chunked loss for a 248K-token vocabulary, and mixed-precision details that keep peak memory at 7.4 GB (Sec. V).
- **Empirical findings, including negative results**: short solutions damage an always-thinking 1.2B model; self-distillation preserves but does not improve it; the same solutions fix truncation for a non-thinking 2B model; and scaling the learned LoRA update by one half keeps the physics gain while reducing the maths regression (Sec. VI).
- **An evaluation protocol for small budgets**: a hashed development/test split of UGPhysics, subject-weighted accuracy, and paired statistics, with a held-out test used exactly once. On it, our model improves accuracy from 15.9% to 18.9% (Sec. VII).

All code, notebooks, the training data, every generated answer, and the model are publicly available [3].

II. Related Work

Physics reasoning benchmarks. Physics appears in broad suites such as MMLU [4] and GPQA [5], and in dedicated benchmarks: SciBench [6] for college-level science, OlympiadBench [7] and JEEBench [8] for competition and

entrance-exam problems, PhysicsEval [9] for free-text solutions graded with a rubric, and UGPhysics [1], which covers 13 undergraduate subjects with typed answers (numbers, expressions, equations, intervals, multiple choice, true/false) and a two-stage judge (MARJ) that combines rule-based matching with a GPT-4o check. We adopt UGPhysics as the primary benchmark and use MATH-500 [10], [11] as a regression check for general mathematical skill.

Science post-training data and reasoning distillation. MegaScience [12] assembles textbook-derived science questions and reference solutions together with other public sets, including physics questions from NaturalReasoning [13]. Distilling long chains of thought [14] from large reasoning models produces strong small models [2], but Li et al. [15] report a learnability gap: models of about 3B parameters or fewer often learn better from shorter chains of thought or from smaller teachers. Long reasoning traces also waste tokens on easy problems [16], and test-time budget control interacts strongly with accuracy [17]. Our results add a related observation: what matters is whether the supervision matches the student's native reasoning mode.

Parameter-efficient fine-tuning. LoRA [18] learns a low-rank update to frozen weights; DoRA [19] decomposes the update into magnitude and direction; QLoRA [20] trains adapters on quantized weights. LoRA tends to forget less of the base model's ability than full fine-tuning while also learning less [21], and forgetting during fine-tuning is well documented [22].

Self-improvement. STaR [23] and rejection-sampling fine-tuning [24] train a model on its own correct solutions. Context distillation [25], [26] trains a model to behave under one prompt as it does under another. We apply both ideas in our self-distillation experiment.

Weight interpolation. Interpolating between pre-trained and fine-tuned weights (WiSE-FT) [27] improves robustness, and task vectors can be scaled and added [28]. Scaling a LoRA update by $\lambda < 1$ is a special case of both, and is what our released model uses.

Benchmark contamination. Overlap with test data inflates scores [29]. N-gram overlap checks, such as the 13-gram test used for GPT-3 [30], miss paraphrases [31]; we therefore combine n-gram, TF-IDF and shared-number criteria.

III. Evaluation Protocol

We designed the evaluation before training, because a small budget invites over-fitting to a small development set.

A. Benchmarks and Generation

UGPhysics (English) contains 5,520 problems in 13 subjects [1]. We use its official zero-shot prompt (a system message asking for step-by-step reasoning and a boxed final answer, followed by the official answer-type template) and its rule-based judge from a pinned commit of the authors' repository, which compares extracted `\boxed{}` answers with the reference as numbers (1% relative tolerance), expressions, equations, intervals, choices or truth values; multi-part questions require every part. **MATH-500** [11] serves as a regression check, graded with the same rule judge. All runs use greedy decoding, at most 4,096 new tokens (the UGPhysics budget for non-o1 models), float16 on a T4 with Hugging Face Transformers [32], and batches of 16 with left padding; before each run we verify that padded batches reproduce single-prompt outputs. A detector stops generations that repeat the same block, which then count as wrong. Only text after a thinking section is graded.

Deviation from the official protocol. MARJ re-checks rule-judge failures with GPT-4o. We did not use this paid second stage, so our absolute accuracies are lower than, and not comparable with, published UGPhysics results; all comparisons in this paper use the same judge.

B. Development and Test Slices

Within each subject, every question receives a fixed rank from the SHA-1 hash of a seed and its identifier, so each rank range is a random sample of its subject. Ranks 0–99 form the *development slice* (1,258 questions; GeometricalOptics has only 58), ranks 0–19 a 260-question *screening subset* for quick checkpoint comparisons, and ranks 100–199 the *held-out test* (1,142 questions in 12 subjects). The test was used once, for the final comparison. The remaining 3,120 questions are untouched. MATH-500 uses the same scheme (10 or 30 problems per subject).

C. Metrics and Statistics

Let S be the subjects present in a slice, N_s the size of subject s in the full English set, and $c_{s,i} \in \{0, 1\}$ the correctness of question i among the n_s evaluated in subject s. We report the subject-weighted accuracy

$$\mathrm{Acc}_w = \sum_{s \in S} \frac{N_s}{\sum_{t \in S} N_t} \frac{1}{n_s} \sum_{i=1}^{n_s} c_{s,i}, \tag{1}$$

which estimates accuracy on the full benchmark from a stratified sample. Intervals are 95% stratified bootstrap intervals (2,000 resamples within subjects) [33]. Two models are compared only on questions both answered: we report the paired bootstrap interval of the difference and the exact McNemar test [34], [35] on discordant questions. With b questions solved only by model A and c only by model B,

$$p = \min\Big(1,\ 2 \sum_{k=0}^{\min(b,c)} \binom{b+c}{k} 2^{-(b+c)}\Big). \tag{2}$$

IV. Training Data

A. Sources and Filtering

Candidates come from MegaScience [12] (textbook-derived physics, physics questions from NaturalReasoning [13], Nemotron-Science multiple-choice physics, and maths) and the PhysicsEval training split [9]. Because the NaturalReasoning and Nemotron pools mix subjects, a multinomial naive Bayes classifier trained on MegaScience's subject labels selects physics; in 5-fold cross-validation it keeps 88.1% of physics questions, while 34.1% of chemistry and 17.1% of

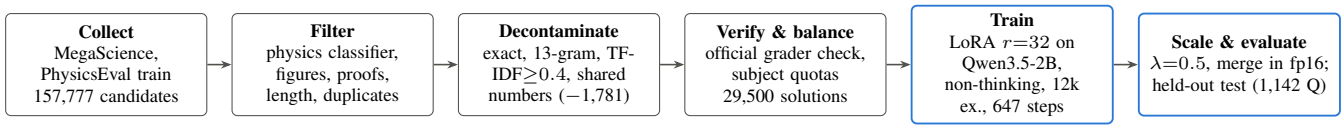


Fig. 1. Overview. The data pipeline (gray) produces 29,500 short, verified physics solutions; the model pipeline (blue) trains a LoRA adapter on a single free T4, applies it at half strength, and evaluates once on held-out questions.

mathematics questions are also accepted, an acceptable trade-off because the later grader check removes unusable items. It kept 90,154 of 100,000 NaturalReasoning and 5,096 of 6,000 Nemotron candidates. We then removed questions that need a figure, table or handbook value (1,305), proofs (1,932), duplicates (551), references too long to check (2,570), maths problems without a boxable answer (2,688), and solutions longer than 2,048 tokens in total or 1,600 tokens of reasoning (23,843).

B. Decontamination

Every candidate was compared with every question of UGPhysics (all 5,520), MATH-500 (all 500) and the PhysicsEval test split. A candidate is removed if (i) its normalized text equals a test question, (ii) it shares any run of 13 words with one [30], (iii) its TF-IDF cosine similarity [36] to one is at least 0.4, or (iv) the similarity is at least 0.3 and at least two uncommon numbers coincide, which catches reworded copies of the same exercise [31]. This removed 1,781 of 157,777 candidates (Table I).

C. Answer Verification and Balancing

Each solution's own boxed answer is compared with its reference answer by the official UGPhysics grader, after normalizing LaTeX formatting. Of 43,170 checkable candidates, 61.7% agreed; disagreeing solutions were dropped, which removes many incorrect or mismatched reference solutions. Solutions with free-form references that cannot be checked automatically were allowed. To spend the budget where the model is weak, a second classifier tags each question with a UGPhysics subject (73.4% accuracy in 5-fold cross-validation on the development questions' subject labels, versus 44.9% for keyword rules; no benchmark text enters the training data), and each subject's quota is its share of the exam multiplied by a weakness weight derived from the baseline's per-subject accuracy, with a 2% floor.

D. Final Dataset

The final set (*v1.2*) has 29,500 training and 500 validation examples (Table I). Each example is a chat prompt, a step-by-step solution and the line "So the final answer is `\boxed{...}`". Solutions are deliberately short: the median example has 763 tokens (LFM2.5 tokenizer; prompt 194, reasoning 479, answer 71), compared with a median of 4,096 tokens (the limit) for the first baseline's answers. The mix is 71.9% exam-style questions with the official prompt, 12.0% plain questions, 10.0% maths, 5.0% short knowledge questions and 1.1% PhysicsEval.

TABLE I
TRAINING DATA (V1.2): SOURCES AND PIPELINE STATISTICS

Source (training split)	Examples	Share
MegaScience, textbook physics	17,548	59.5%
MegaScience, NaturalReasoning physics	7,390	25.1%
MegaScience, maths	2,943	10.0%
MegaScience, Nemotron-Science physics (MC)	1,298	4.4%
PhysicsEval, training split	321	1.1%
Total	**29,500**	
Decontamination (of 157,777 candidates)		
exact match / 13-word run	110 / 845	
TF-IDF $\geq$ 0.4 / shared numbers	792 / 34	
Grader check (43,170 checkable candidates)		
agree / agree after normalizing	15,999 / 10,636	
disagree (dropped)	16,535	

V. MODELS AND TRAINING

A. Base Models

Qwen3.5-2B [37] is a hybrid model whose text component (about 1.9B parameters; the vision encoder is not used) has 24 layers: 18 Gated DeltaNet linear-attention layers [38] and 6 full softmax-attention layers, hidden size 2,048, and a 248,320-token vocabulary. Its chat template offers a thinking mode and a non-thinking mode in which the assistant turn begins with an empty thinking block; it follows the Qwen3 family [39]. **LFM2.5-1.2B-Thinking** [40] is a 1.2B hybrid model that always reasons inside a thinking section before answering. Qwen3.5-2B is released under the Apache-2.0 license.

B. Low-Rank Adaptation

For a frozen weight $W \in \mathbb{R}^{d\times k}$, LoRA [18] learns $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$ and uses

$$W' = W + \lambda \frac{\alpha}{r} BA, \tag{3}$$

with $\lambda = 1$ during training. On Qwen3.5-2B we adapt every large linear map: the query, key, value and output projections of the 6 attention layers; the fused QKV input projection, the output-gate projection and the output projection of the 18 Gated DeltaNet layers; and the gate, up and down projections of all 24 MLPs (150 matrices). The small per-head projections of the delta rule are left frozen. With $r = 32$ and $\alpha = 64$ this gives 31.26M trainable parameters (1.63%). The loss is the token-level cross-entropy over the solution, answer and end-of-turn tokens only, averaged over all such tokens in an optimizer step.

Format. For the non-thinking model, prompts are rendered exactly as at evaluation (non-thinking chat template; system

TABLE II
TRAINING CONFIGURATION OF THE RELEASED MODEL

Base model	Qwen3.5-2B, text model, non-thinking format
Adapter	LoRA, $r = 32$, $\alpha = 64$ (released with $\alpha = 32$)
Trainable parameters	31.26M (1.63%), 150 matrices
Data	12,000 random v1.2 examples (seed 3407) 11,975 within 2,048 tokens
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 0.01
Learning rate	10^{-4}, 5% linear warm-up, cosine to 10%
Batch	$\leq$8,192 tokens per pass, $\approx$18.5 examples per step
Steps	647 (one epoch), gradient clipping at 1.0
Precision	fp16 weights, fp32 adapters, fp16 autocast
Memory, time	7.4 GB peak, about 4.5 h on one T4
Validation loss	0.2949 at the final step (200 examples)

messages that name another model's developer are dropped), and the target is the solution as plain text followed by the boxed answer. For the thinking model, the solution is placed inside the thinking section, copying the delimiters from the model's own outputs.

Adapter scaling. After training, the released model uses $\lambda = 0.5$ in (3), implemented by setting $\alpha = 32$ in the adapter configuration without retraining. This is a linear interpolation between the base and fine-tuned weights, as in WiSE-FT [27] and task arithmetic [28].

C. Fitting a Hybrid Model on a T4

The T4 lacks bfloat16, so the frozen base weights are stored in float16, the adapters in float32, and the forward pass runs under float16 autocast with dynamic loss scaling [41]; only the first 5–6 steps overflowed while the loss scale settled. Gradient checkpointing [42] is required: without it, a single 2,048-token forward pass exceeded the T4's memory. The output layer is the largest consumer of transient memory, because the vocabulary has $V = 248{,}320$ entries. We compute logits only at trained positions and in chunks of $\lfloor 2^{26}/V \rfloor = 270$ positions, each wrapped in a recomputation checkpoint so that no more than about 67M logits exist at once. Examples are sorted by length within shuffled groups of 512 and packed into forward passes of at most 8,192 tokens and 8 examples, giving about 18.5 examples per optimizer step. The Gated DeltaNet layers run on the reference PyTorch implementation of the chunked delta rule, as the optimized kernels were not installed. Peak memory was 7.4 of 14.6 GB and a step took 22–33 s. Table II lists the configuration.

D. Release Precision

The adapter is merged into the base weights in float32 and saved in float16, the precision used in every evaluation. A first merge saved in bfloat16 changed 9 of 26 greedy answers on a check set (5 versus 6 correct): bfloat16's 8-bit significand has a rounding step of about 1.2×10^{-4} for weights of magnitude 0.02, comparable to many entries of the scaled LoRA update, so part of the update is rounded away. The adapter itself is released as well.

TABLE III
ALL EXPERIMENTS ON THE SCREENING SUBSETS (UGPHYSICS: 260 QUESTIONS; MATH-500: 70). ACCURACY IN %, WEIGHTED BY SUBJECT.

Model / training	**UGPh.**	**MATH**	**Trunc.**
LFM2.5-1.2B-Thinking, 4,096 tokens	13.3	67.5	61%[a]
LFM2.5-1.2B-Thinking, 8,192 tokens	14.5	–	–
+ short solutions (Exp. 1), step 600	5.5	46.8	3%
+ self-distillation (Exp. 2), step 293	10.9	66.5	71%
Qwen3.5-2B, thinking mode	0.0	–	74%[b]
Qwen3.5-2B, non-thinking	17.5	67.1	52%[a]
+ v1.2 LoRA (Exp. 3), step 250	16.7	53.7	3%
+ v1.2 LoRA (Exp. 3), step 500	16.1	58.8	1%
+ v1.2 LoRA (Exp. 3), step 647	**19.0**	51.7	2%
+ half strength (Exp. 4), released	17.3	**69.4**	7%

Trunc.: share of UGPhysics answers that hit the token limit. [a]Measured on the full development slice. [b]The remaining 26% were stopped as repetition loops; no answer left the thinking section (672 questions).

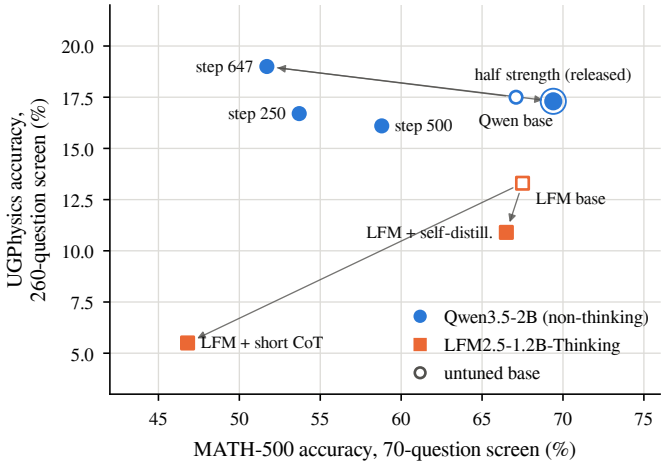


Fig. 2. Physics versus maths accuracy on the screening subsets. Arrows connect each base model to its fine-tuned variants. Screening estimates have wide intervals (about ± 5 points); larger evaluations follow in Sec. VII.

VI. EXPERIMENTS

Table III summarizes all experiments on the screening subsets; Fig. 2 shows the same results as a trade-off between physics and maths.

A. Baselines

LFM2.5-1.2B-Thinking scores 11.3% [9.1, 13.5] on the 1,258 development questions; 61% of its answers reach the 4,096-token limit while still thinking. Doubling the budget raises its screening accuracy only from 13.3% to 14.5%, so the budget is not its main limitation. Qwen3.5-2B in non-thinking mode scores 14.5% [11.9, 17.3] on 1,008 development questions, but 51.9% of its answers reach the limit and 41.8% contain no boxed answer: it rambles rather than reasons. In thinking mode, under greedy decoding and the same budget, none of its 672 answers left the thinking section, so it scores 0%. This mode may perform far better with sampling and a much larger budget, which we did not test.

B. Experiment 1: Short Solutions for a Thinking Model

We first trained LFM2.5-1.2B-Thinking with LoRA on v1.2 (solutions inside its thinking section; 600 steps of 16 examples). DoRA [19] was 1.75× slower per step on the T4, so LoRA was used. The format was learned (97% of answers finished), but UGPhysics accuracy fell from 13.3% to 5.5% and MATH-500 from 67.5% to 46.8%. Replacing a model's long native reasoning with 500-token solutions removed the reasoning it relies on.

C. Experiment 2: Self-Distillation

To teach the thinking model to finish without changing its reasoning style, we sampled two answers for each of 9,000 training questions (temperature 0.6, top-p 0.95, up to 3,584 tokens) with vLLM [43], graded them with the official judge, and kept the shortest correct answer per question [23], [24]. Only 15.0% of answers were correct and only 7.9% of questions received one, mostly in mathematics; correct answers to physics questions were rare (about 1%). A first round produced answers that argued with the prompt about the answer format, so the second round generated with a plain prompt and trained under the official prompt (context distillation [25], [26]), with filters against format talk and text after the final answer (2,437 examples). After 293 steps, MATH-500 was preserved (66.5%), but UGPhysics was not improved (10.9%) and 71% of answers still reached the limit. With so few correct physics solutions, there was little skill to reinforce.

D. Experiment 3: Non-Thinking Qwen3.5-2B

Qwen3.5-2B in non-thinking mode writes visible step-by-step reasoning in the same style as v1.2, and its main weakness is format, so we trained it on v1.2 (Sec. V). Truncation fell from 52% to 2% and the median answer shrank to 516 tokens. The final checkpoint reached 19.0% on the screening subset and 16.5% [14.0, 18.9] on the full development slice; on the 1,008 questions both models answered, the gain over the base model was +2.0 points $[-0.8, 5.1]$ ($p = 0.071$). MATH-500 fell from 67.1% to 51.7%, and from 40% to 13% on its hardest level.

E. Experiment 4: Half-Strength Adapter

Applying the same adapter with $\lambda = 0.5$ recovered MATH-500 on the screening subset (69.4%) and its hardest level (33%), while answers lengthened moderately (median 710 tokens; 7.7% truncated). On the development slice it reached 18.3% [15.6, 20.9]: +3.3 points [0.4, 6.4] over the base model on their 1,008 common questions (86 versus 51 discordant wins, $p = 0.0035$) and +1.8 points $[-0.4, 4.1]$ over the full-strength adapter ($p = 0.095$). It also outperformed LFM2.5-1.2B-Thinking on all 1,258 questions by 6.9 points [4.6, 9.3] ($p < 10^{-4}$). We selected this model for the held-out test. The value $\lambda = 0.5$ was fixed in advance; no other values were tried.

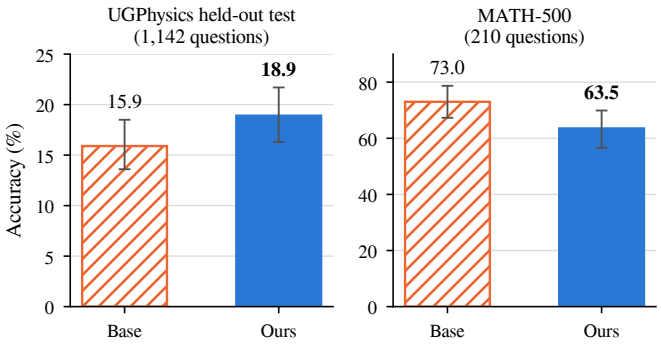


Fig. 3. Held-out UGPhysics and MATH-500 accuracy of the base model and ours. Whiskers: 95% stratified bootstrap intervals.

VII. Held-Out Results

A. Main Result

On the 1,142 held-out questions, our model reaches 18.9% [16.3, 21.7] against 15.9% [13.6, 18.5] for the base model (Table IV, Fig. 3). The paired difference is +3.0 points [0.3, 5.7]; our model solves 102 questions the base model misses and misses 75 that it solves (exact McNemar $p = 0.050$). Unweighted accuracy shows the same gap (18.0% versus 15.7%). The effect closely matches the development estimate (+3.3), which suggests that selecting the model on the development slice did not inflate it noticeably. On MATH-500 (210 problems), however, our model is 9.4 points worse (63.5% versus 73.0%; $p = 0.044$), mostly at Level 5 (26.4% versus 49.1%) and Level 4 (61.0% versus 72.9%), while Levels 1 and 3 improved (88.2% versus 70.6%; 78.6% versus 71.4%).

B. Where the Gain Comes From

Fig. 4 breaks the held-out result down by subject. The largest gains are in Classical Mechanics (+10.0), Semiconductor Physics (+8.1), Electrodynamics (+7.2) and Quantum Mechanics (+7.0); Wave Optics (-3.0) and Statistical Mechanics (-2.0) declined slightly. With 72–100 questions per subject, individual differences below about 10 points are within noise. By answer type (Table V), the largest change is in multiple choice (25.7% to 68.6%, 35 questions) and equations (9.5% to 18.2%), while true/false questions declined (68.8% to 56.2%, 32 questions). By reasoning skill, knowledge recall gained most (32.4% to 41.9%), whereas mathematical derivation, the hardest category, improved only from 6.8% to 9.4%.

C. Generation Behavior and Cost

Table VI and Fig. 5 compare how answers end. The base model finishes only 48.9% of its held-out answers and loses the same share to the token limit; ours finishes 91.7%, with a mean length of 1,006 tokens versus 2,720. Evaluating the held-out test took 4.71 T4-hours for our model versus 6.31 for the base model.

TABLE IV
MAIN RESULTS. ACCURACY (%) WITH 95% INTERVALS; DIFFERENCES ARE OURS MINUS BASE ON QUESTIONS BOTH ANSWERED.

Evaluation	Questions	Qwen3.5-2B (base)	Ours	Difference	Wins / losses	McNemar p
UGPhysics, held-out test (ranks 100–199)	1,142	15.9 [13.6, 18.5]	**18.9** [16.3, 21.7]	**+3.0** [+0.3, +5.7]	102 / 75	0.050
UGPhysics, development (ranks 0–99)	1,008	14.5 [11.9, 17.3]	**17.9**	+3.3 [+0.4, +6.4]	86 / 51	0.0035
MATH-500 (regression check)	210	**73.0** [67.3, 78.7]	63.5 [56.6, 69.9]	−9.4 [−16.5, −2.6]	20 / 36	0.044

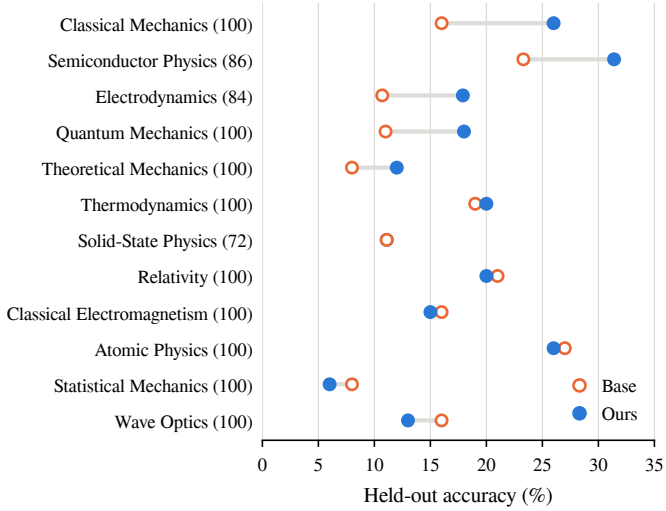


Fig. 4. Held-out accuracy by subject (questions per subject in parentheses), sorted by the change. Solid-State Physics is unchanged (11.1%).

TABLE V
HELD-OUT ACCURACY (%) BY ANSWER TYPE AND REASONING SKILL. THREE QUESTIONS WITH MALFORMED TYPE LABELS AND 40 WITHOUT A SKILL LABEL ARE OMITTED.

Category	n	Base	Ours
Numerical value	384	24.2	25.8
Expression	304	7.6	6.6
Equation	148	9.5	18.2
Multiple choice	35	25.7	68.6
True/false	32	68.8	56.2
Interval	12	8.3	0.0
Several answers	224	7.1	7.1
Knowledge recall	105	32.4	41.9
Laws application	412	20.9	21.6
Math derivation	468	6.8	9.4
Practical application	117	18.8	20.5

VIII. DISCUSSION

Truncation is not the main obstacle; competence is. Fine-tuning eliminated most truncation (49% to 7%), yet accuracy rose by only three points. Most questions the base model failed to finish would have been answered incorrectly anyway, as the LFM budget experiment (13.3% at 4,096 tokens, 14.5% at 8,192) also suggests. At this scale, short supervised solutions mainly teach the model to reach and state an answer; they add little physics knowledge.

Match the supervision to the student's reasoning mode. The same data hurt the always-thinking model (13.3% to 5.5%) and helped the non-thinking one. Li et al. [15] found that small students learn better from short chains of thought than from long ones; our results suggest a complementary condition: short supervision works when the student already reasons in that register, and harms a student whose competence depends on long internal deliberation.

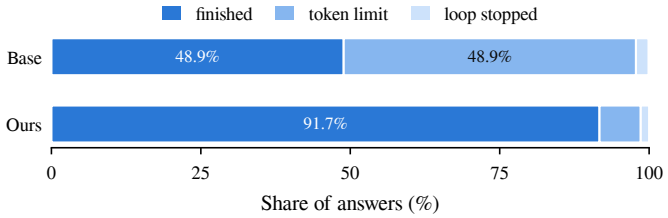


Fig. 5. How held-out answers end: finished normally, cut at the 4,096-token limit, or stopped by the repetition detector.

TABLE VI
GENERATION BEHAVIOR ON THE HELD-OUT TEST (1,142 QUESTIONS)

	Base	Ours
Mean / median new tokens	2,720 / 3,862	1,006 / 726
90th percentile of new tokens	4,096	1,782
Finished normally	48.9%	91.7%
Hit the 4,096-token limit	48.9%	6.9%
Stopped as a repetition loop	2.1%	1.4%
No boxed answer	38.5%	8.1%
T4 GPU-hours for the test	6.31	4.71

Half strength trades little physics for much maths. Halving the update kept the physics gain (on the development slice it was even 1.8 points higher, not significantly) while sharply reducing the maths regression on the screening subset, consistent with interpolation results in other domains [27], [28]. A plausible reading of Table V is that much of the gain reflects answer-format behavior (choosing and stating a single option, writing an equation in the expected form), which survives scaling, while the brevity that hurts hard maths problems is partly undone. The large multiple-choice gain (35 questions) supports this reading, but the subsets are small.

Small screens mislead. On 70 MATH-500 problems the released model looked slightly better than the base model (+2.3); on 210 it was 9.4 points worse. Development decisions based on 260 or 70 questions carry intervals of about ±5 points. Keeping a held-out slice for one final comparison, and reporting paired statistics, was essential to an honest result.

Precision is part of the model. A LoRA update is small relative to the weights it modifies. Saving merged weights

in a format with a coarser significand than the one used in evaluation measurably changed outputs. Evaluating and releasing in the same precision, and releasing the adapter itself, avoids this.

IX. Limitations

Our judge is the rule-based stage of UGPhysics only; the official protocol would credit some answers our judge rejects, so absolute numbers are not comparable with published results. All results come from a single training run and seed. The held-out improvement is supported by a paired bootstrap interval that excludes zero, while the exact McNemar test is at the 5% boundary ($p = 0.050$); the development slice gives independent supporting evidence ($p = 0.0035$). We compared against two model families only; other small models, such as Qwen3-1.7B, DeepSeek-R1-Distill-Qwen-1.5B [2], or MegaScience's own fine-tunes [12], were not evaluated with our harness, and Qwen3.5-2B's thinking mode was only evaluated under greedy decoding and a 4,096-token budget, which does not represent its best configuration. The model is worse than its base on hard mathematics. Decontamination cannot exclude paraphrases that share neither wording nor numbers with a test question. Finally, the subject tagger was trained on the development questions' subject labels; it influenced only how many examples each subject received.

X. Ethics and Broader Impact

The model answers about four in five undergraduate physics questions incorrectly and should not be used to grade students or as an authority on physics; any educational use should present its answers as drafts to be checked. Lowering the cost of domain adaptation makes such work accessible to students and to institutions without dedicated hardware. Most training data derives from MegaScience, which is licensed CC BY-NC-SA 4.0; the released data follows that license, and users of the model should consider it.

XI. Reproducibility

The repository [3] contains the evaluation harness, the data builder, the trainer, and the Colab notebooks for every stage; the v1.2 training data; the training configurations and logs; every generated and graded answer of every evaluation run (question identifiers only, not benchmark text); and the reports from which all numbers in this paper are taken. The model, in float16, and the half-strength adapter are published on Hugging Face (huggingface.co/eswaroy/qwen3.5-2b-physics). All experiments ran on free-tier NVIDIA T4 GPUs.

XII. Conclusion

With leak-checked, grader-verified short solutions, a LoRA adapter trained in 4.5 GPU-hours on a free T4, and the update applied at half strength, a 2B model improved from 15.9% to 18.9% on held-out undergraduate physics questions, finished 92% instead of 49% of its answers, and became cheaper to run, at the cost of 9.4 points on competition mathematics. The negative results are as instructive as the positive one: supervision must match the student's reasoning mode, self-distillation cannot create skill the model lacks, and small development screens are unreliable without a held-out test. Closing the remaining gap will likely require verified long-form solutions from stronger teachers and training that protects the base model's mathematics.

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